Victoria Barnett

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PROFESSIONAL SUMMARY

Software and ML practitioner with a background spanning data collection operations, deep learning instruction, and Air Force QA. Turned a failing offshore vendor into the leading producer and became the case study for a quality-review framework taught org-wide. Comfortable working close to hardware, data pipelines, and production systems. Portfolio, project writeups, and course material at vb4r.com.

EXPERIENCE

Lecturer and Course Developer May 2026 – Present
San Francisco State University *San Francisco, CA*

- Progressed from instructional assistant to course developer to lecturer for CSC 511: Deep Learning; current term ends December 2026.
- Designed an intermediate, interdisciplinary machine learning course; teaching its first offering to 30 students in the PINC computing applications minor, anchoring ML heuristics in domain-specific examples across the range of majors in class.
- Students implement gradient descent, backpropagation, and attention from scratch, then build a working transformer encoder in PyTorch and Lightning.

Data Coordinator, Shift Supervisor February 2026 – August 2026
Ascen Workforce, LLC *San Francisco, CA*

- Ran collection operations for a confidential client: managed a fleet of collection devices, dozens of operators, and a shift structure that grew from a single pilot plan into multi-site production.
- Served as one of three coordinators reporting to the R&D lead for a new hardware platform, carried alongside the standard coordinator role.
- Collaborated with a staff ML engineer to align captures and narrow the gap between two modalities using a mixed strategy that generalized sufficiently to retarget cheaper captures at scale.
- Queried an undocumented schema with LLMs to build SQL and Retool views tracking product and experiment quality across sites; traced the majority of logged device faults to an idle-timeout error guard blocking between clips and developed an alternative procedure with no per-instance downtime growth, clarifying telemetry.
- During a vendor replacement, rebuilt task instructions as form-based objects, implemented rapid task-swapping logic from operator hardware, and replaced leaderboard CTEs with functional logic for near-real-time statistics.

Instructional Assistant December 2025 – May 2026
San Francisco State University *San Francisco, CA*

- Led twice-weekly recitations for approximately 40 life-science students in CSC 511: Protein Modeling with Deep Learning, covering AlphaFold2 architectures.
- Wrote tutorial notebooks on model architectures, linear algebra operations, and Lightning; led study guide recitations.
- Diagnosed Colab compute issues, Python environment errors, and model evaluation problems with students.

Chapter Co-Chair August 2023 – August 2025
Democratic Socialists of America *San Francisco, CA*

- Elected co-chair of a 2,300-member chapter; facilitated large meetings under Robert's Rules and planned strategic campaigns. As administrative co-chair, migrated chapter docs to Outline, put three tools behind Auth0 login, and maintained fifteen Docker (Portainer) services.

IT Consultant September 2023 – August 2024
Hacknsleep *San Francisco, CA*

- Repaired Eero and Google mesh networks and Sonic hookups across sublets; recommended the most cost-effective fix per site and coordinated contract labor for larger installs.

Crew Chief, then Quality Assurance Inspector November 2016 – November 2021
U.S. Air Force *Eglin AFB, FL*

- Directed 10 contractors and 2 airmen through 3,000 munitions test and 670 maintenance operations; awarded a signed Special Experience Identifier for the inspection track.
- Reproduced and documented a defect in a published technical order that could result in a hazardous asset mechanical failure, proposed the correction, and carried it through review to a global safety data revision on assets worth more than $500 million.
- Structured a new Microsoft Access report flow using data from a legacy DBMS interface, FOCUS.

PROJECTS

Recall-First Screening Classifier https://github.com/victoria-riley-barnett/r-alzheimers-rf

- Shifted the classification threshold toward detection, cutting missed diagnoses from 51 to 16 (68% fewer) for a cost of 0.56 points of accuracy.

Tailscale for KOReader https://github.com/victoria-riley-barnett/koreader-tailscale

- Built a Tailscale integration for KOReader, making VPN access easy on embedded e-reader hardware.

SKILLS

Languages: Python, SQL, C, C++, Rust, Java, JavaScript, TypeScript, WebAssembly, Lisp, HTML, CSS
ML and Data: PyTorch, Lightning, Pandas, Retool
Frameworks and Libraries: Django, Node.js, React, MapLibre GL, Three.js
Tools and Infrastructure: Docker, systemd, Tailscale, Auth0

EDUCATION

San Francisco State University San Francisco, CA
B.S., Computer Science. Dean's List, Fall 2022; GI Bill. *December 2025*
Relevant Coursework: Data Structures and Algorithms, Operating Systems, Compilers, Database Systems, Linear Algebra, NLP, AI Explainability and Ethics